

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 02
Vectors.

Review

1. What is a *block* vector?
2. What does $\mathbf{nnz}(\vec{v})$ mean?
3. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$, what is $\mathbf{nnz}(\vec{v})$?

New (from Chp.1)

1. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} + \vec{u}$?

2. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, what is $\vec{v} - \vec{u}$?

3. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$, what is $4\vec{v}$?

4. What is a *linear combination*?

5. If $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, find the linear combination $4\vec{v} - 3\vec{u}$.

6. Find the *inner product* of $\vec{v} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.